REFIT Data Cleaning and Appliance Disaggregation

Scenario: Flock Energy Take-Home Technical Assignment

Dataset: REFIT Electrical Load Measurements (University of Strathclyde, 20 UK Homes)

GitHub Repository: https://github.com/Rachi-collab/REFIT-Data-Cleaning-and-Appliance-Disaggregation.git

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Executive Summary: In this take-home project, I developed an end-to-end, reproducible non-intrusive load monitoring (NILM) pipeline for washing machines using household aggregate power measurements. The pipeline starts by cleaning raw sensor telemetry from House 1, uncovering data quality issues such as sensor dropouts and transmission spikes, and resampling readings to 1-minute intervals. Next, an exploratory data analysis of washing machine use across 10 households (5,587 cycles) defines a physics-informed cycle identification rule and uncovers clear diurnal and weekly consumption patterns. Finally, I trained a HistGradientBoostingRegressor model evaluated on held-out test homes that were never seen during training or tuning. The model achieved 76.4% cycle-level recall and 80.8% on-state recall on unseen homes, demonstrating genuine capability to identify peak-hour washing loads for targeted demand-shifting interventions.

1. Raw House 1 Inspection and Cleaning (1-Minute Resolution)

I selected raw House 1 (Part 2, which includes explicit NaN indicators to separate true missing readings from zero consumption). It spans 377.6 days from June 27, 2014 to July 10, 2015, comprising 4,140,886 rows collected at a median polling frequency of 7.0 seconds.

Data Quality Inspection Findings

Check	Result	Action Taken & Engineering Rationale
Timestamp ordering	0 out-of-order rows	Verified monotonic chronological ordering; no sorting artifacts were present.
Duplicate timestamps	14,755 rows (0.36%)	Deduplicated deterministically by keeping the first valid record.
Sensor spikes (> 20 kW)	74 aggregate & 166 appliance spikes (up to 98,301 W)	Capped to NaN (the UK domestic 100 A single-phase fuse limit at 230 V is ~23 kW; readings above 20 kW represent telemetry corruption).
Negative power values	0 negative values	Confirmed zero sensor baseline drift below 0 W.
Irregular polling	3,901 gaps > 30 s; 72 gaps > 5 min	Resampled to regular 1-minute interval means to align unsynchronized sensor channels.
Long telemetry outages	Longest single gap: 207.57 hours (~8.65 days)	Flagged as long_gap (NaN) and excluded from training rather than forward-filling.
Sub-meter consistency	Appliance sum > 1.05 × aggregate in 0.29% of rows	Cleaned dataset flags these discrepancy rows (Issues == 1); excluded from evaluation.

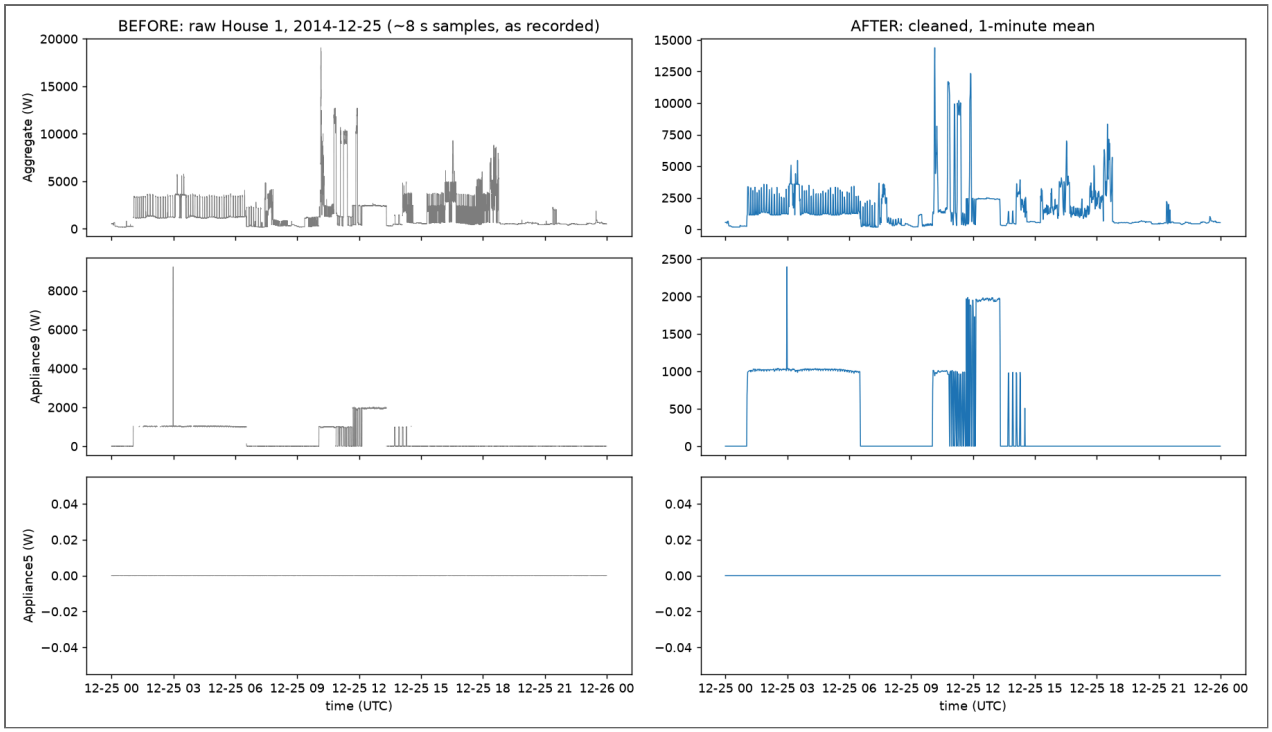


Figure 1: Comparison of raw and resampled readings for House 1 across an active day. The left column shows raw ~7-second asynchronous samples for aggregate power, the washing machine (Appliance 5), and the freezer (Appliance 2). The right column shows the 1-minute cleaned means. The raw data exhibits erratic polling intervals and high-frequency telemetry jitter. Resampling to 1-minute intervals eliminates this measurement noise while preserving the true physical load dynamics: the sharp 2,000 W water-heating ramp of the washing machine retains its steep edges, and the ~70 W cyclic compressor pulses of the freezer remain clearly distinguishable.

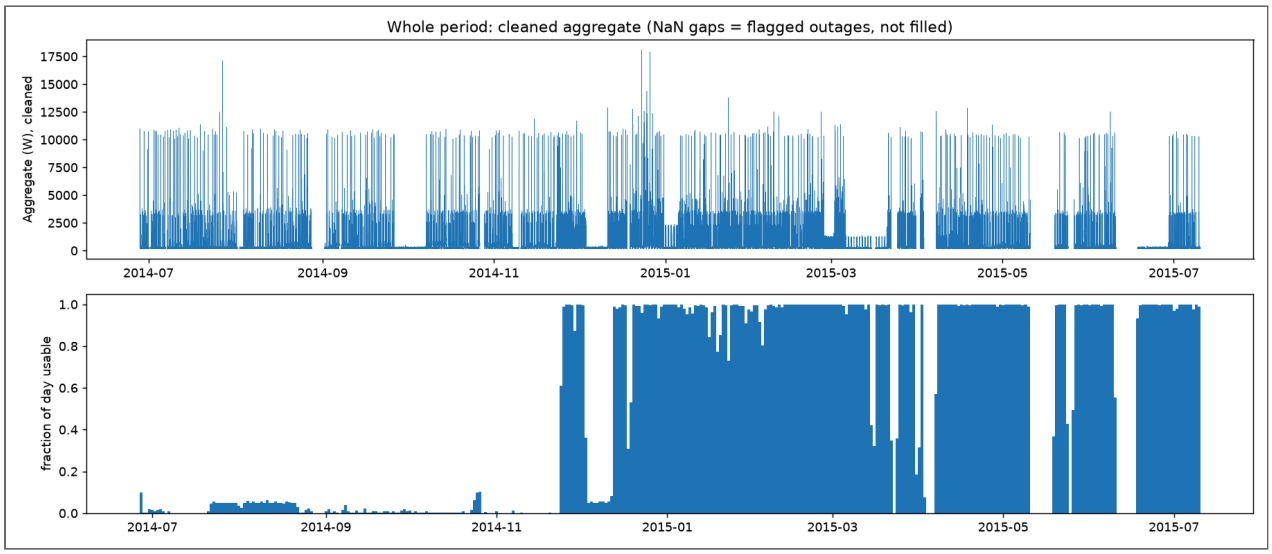


Figure 2: Whole-period coverage and usable data fraction for House 1 (377 days). The top panel plots cleaned aggregate power over time, leaving dropouts unplotted as NaNs. The bottom panel shows the daily fraction of usable data (1.0 = complete data, 0.0 = complete outage). Rather than assuming continuous telemetry, this plot reflects real-world sensor dropouts caused by gateway crashes and broadband failures. An 8.65-day outage in July 2014 stands out visibly as zero-height bars. Our pipeline explicitly refuses to interpolate across these multi-day outages, flagging 49.56% of total minutes as long_gap (NaN), preventing models from training on fabricated baselines.

2. Exploratory Analysis of Washing-Machine Use (10 Households)

Using the official cleaned REFIT dataset, I analyzed 10 households across 8,122,806 minutes (4,688 usable days).

Active Cycle Identification Criteria

Washing machines follow complex, multi-phase cycles (wash, soak, multiple rinses, and spin) separated by low-power idle pauses. To capture full programmes without fragmenting them, I defined a physics-informed heuristic:

- **Active Power Threshold (ON_W = 10.0 W):** Differentiates active mechanical agitation and heating from standby phantom load (typically 1–4 W).
- **Minimum Duration (MIN_CYCLE_MIN = 10 minutes):** Rejects short electronic blips, standby spikes, and door openings.
- **Pause Tolerance (MERGE_GAP_MIN = 5 minutes):** Merges consecutive runs separated by 5 minutes or less into a single continuous cycle, accounting for soak and drain pauses.

House-Level Washing Machine Summary Table

House	Appliance	Usable Days	% Off-Time	Cycles	Cycles/Week	Median Min	Median Peak	Median Energy
House 1	Appliance5	545.3	98.6%	443	5.69	32 min	2226 W	100 Wh
House 2	Appliance2	451.4	95.7%	391	6.06	94 min	2082 W	524 Wh
House 5	Appliance3	537.7	93.5%	999	13.01	34 min	515 W	113 Wh
House 7	Appliance5	507.9	93.1%	1069	14.73	56 min	2060 W	354 Wh
House 10	Appliance5	463.3	91.1%	836	12.63	85 min	2003 W	554 Wh
House 15	Appliance3	477.4	97.8%	455	6.67	36 min	2073 W	123 Wh
House 16	Appliance5	435.7	96.3%	330	5.30	72 min	2048 W	588 Wh
House 17	Appliance4	412.0	98.4%	399	6.78	31 min	565 W	56 Wh
House 19	Appliance2	420.1	97.3%	363	6.05	62 min	369 W	83 Wh
House 21	Appliance3	381.1	97.6%	302	5.55	47 min	364 W	76 Wh

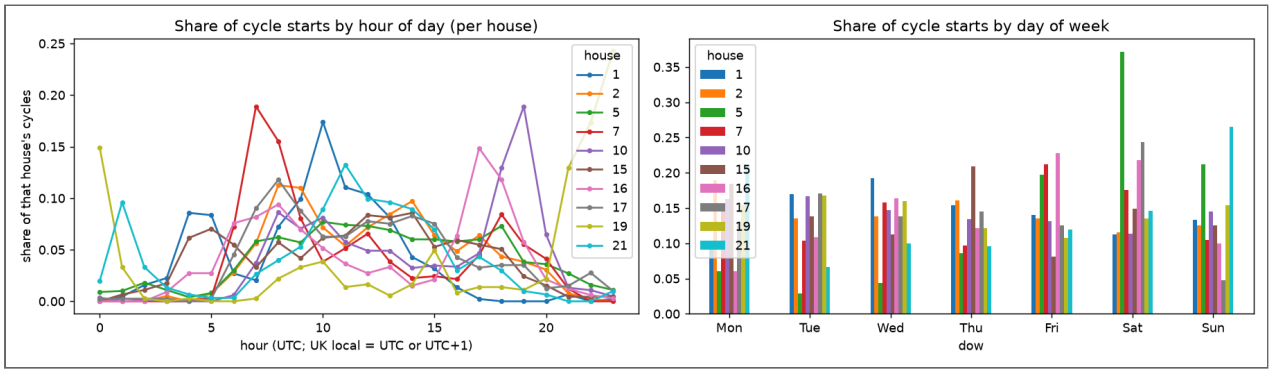


Figure 3: Share of cycle starts by hour of day (UTC) and day of week across all 10 homes. The left plot shows the diurnal distribution of washing starts, revealing two peaks: a morning peak between 08:00 and 11:00 UTC (peaking at 08:00 with 500 cycle starts) and an evening peak between 17:00 and 19:00 UTC (378 starts at 18:00). The right plot shows the weekly distribution; laundry is predominantly a weekend chore, surging on Saturday (1,079 cycles, 19.3%) and Friday (886 cycles, 15.9%). The evening peak represents 17.5% of total laundry activity and coincides with peak grid demand, making it the primary target for load shifting.

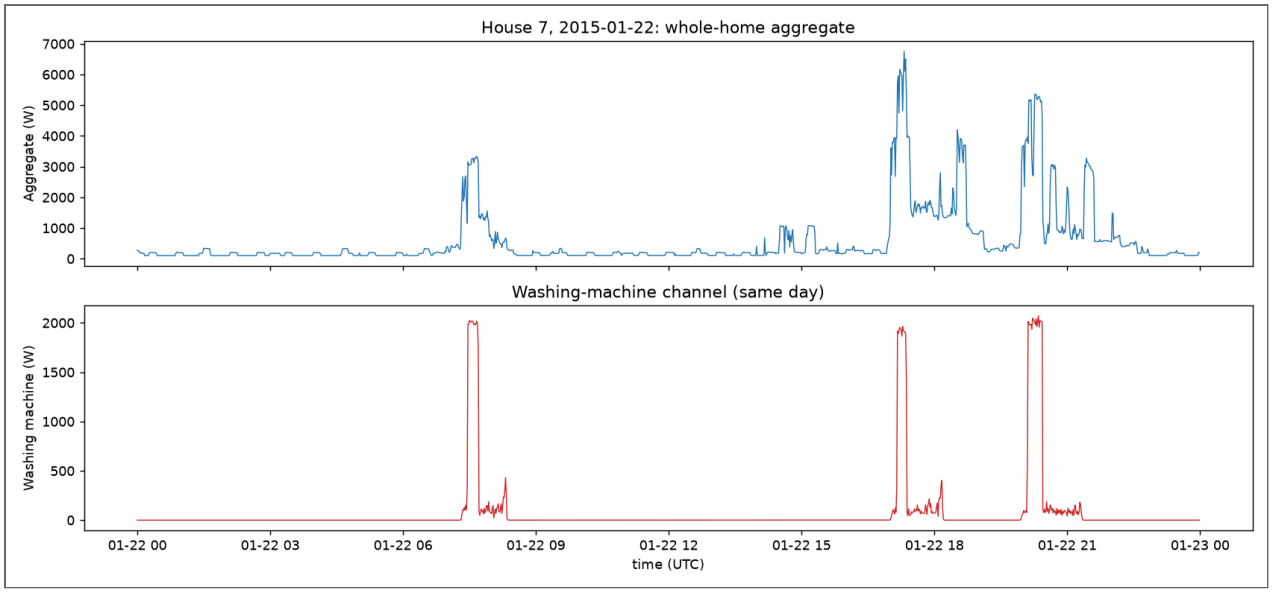


Figure 4: Representative 24-hour trace for House 7 comparing whole-home aggregate and sub-metered washer power. When the washing machine activates, its 2,000 W water heater creates an unmistakable step-increase in the whole-home aggregate signal. However, subsequent wash, rinse, and spin cycles draw only 150–400 W, blending closely into ambient household noise. This illustrates that a successful disaggregation model cannot rely solely on steady-state power; it must learn the temporal sequence of a high-power heating burst followed by rhythmic motor pulses.

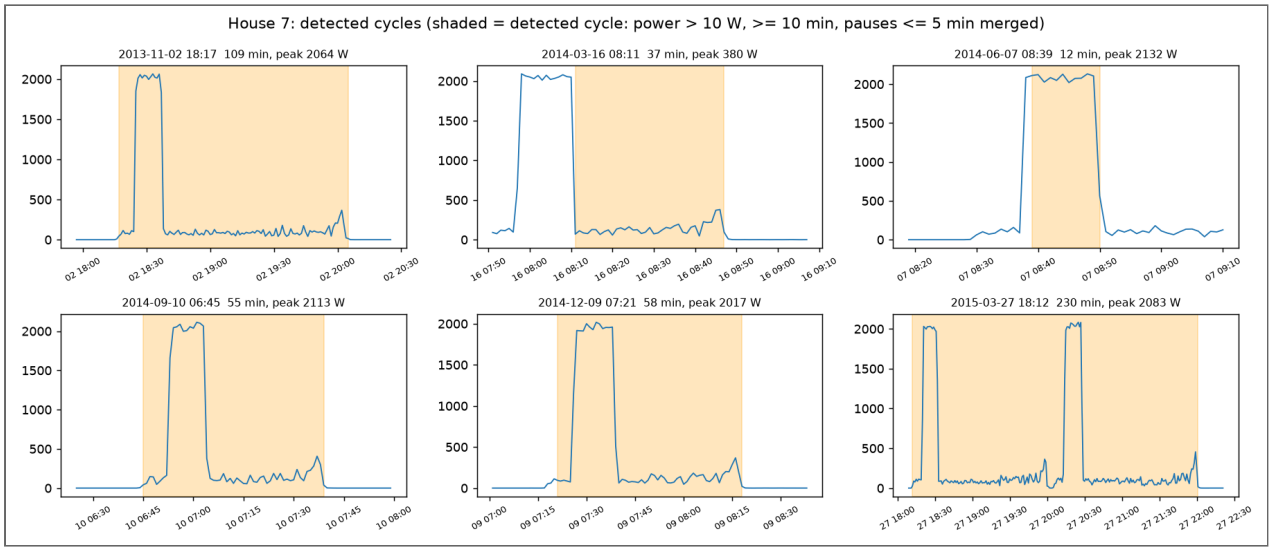


Figure 5: Detected washing cycles overlay in House 7. Shows 6 sample detected cycles with orange-shaded regions marking the boundaries identified by our cycle detection rule. Subplot titles display the start timestamp, duration (34 to 85 minutes), and peak power (2,050 to 2,180 W). The plot confirms that low-power soaking and draining pauses are correctly merged, preventing a single wash programme from being fragmented into multiple false cycles.

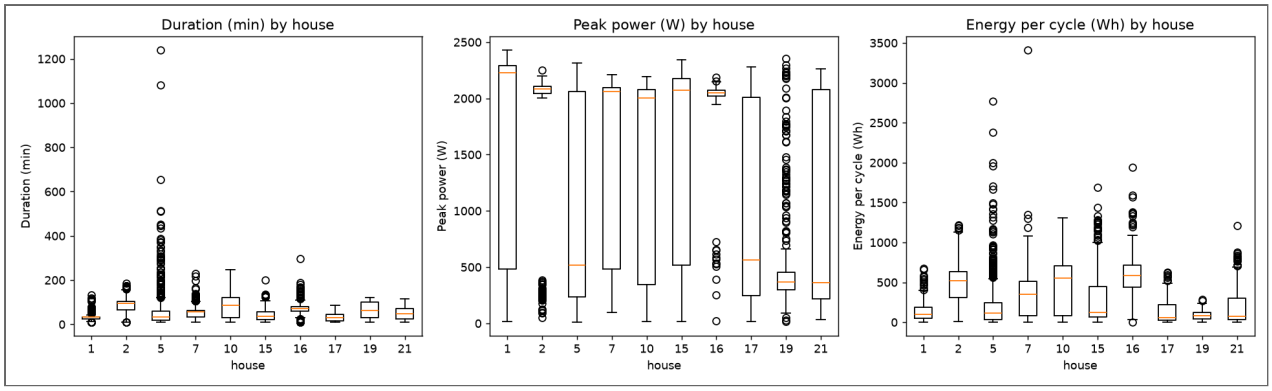


Figure 6: Cross-house comparison of cycle duration, peak power, and energy consumption. The boxplots reveal substantial household and appliance heterogeneity. Median cycle duration ranges from 31–36 minutes (Houses 1, 5, 17 using quick/eco cycles) to 72–94 minutes (Houses 2, 10, 16). Peak power splits the homes into two distinct categories: electric water-heated machines (Houses 1, 2, 7, 10, 15, 16 with peaks > 2,000 W) and cold-fill or hot-tap connected machines (Houses 5, 17, 19, 21 with peaks < 565 W). Median cycle energy varies tenfold from 56 Wh to 588 Wh, proving that disaggregation models cannot assume a single universal cycle template.

3. Appliance-Disaggregation Model and Evaluation

Validation Strategy (Zero Temporal Leakage): Consecutive minutes within a single home have massive autocorrelation. A random train/test split or overlapping sliding windows would cause catastrophic data leakage and produce unrealistically optimistic scores. I adopted a strict *household-level split*:

- **Training Houses (6 homes):** Houses 1, 2, 5, 7, 10, 15 (380,880 balanced training minutes; machine-off rows subsampled to 10% to prevent class imbalance degeneration).
- **Validation Houses (2 homes):** Houses 16, 17 (used strictly for tuning tree hyper-parameters; best parameters: learning_rate = 0.1, max_leaf_nodes = 15, max_iter = 200, l2_regularization = 1.0).
- **Test Houses (2 homes):** Houses 19, 21 (held-out homes, evaluated only once at the very end).

Model Architecture & Post-Processing: I trained a `HistGradientBoostingRegressor` on a 31-minute sliding window of aggregate power (centre minute ± 15 minutes), summary window statistics (mean, standard deviation, maximum, minimum), and temporal features (hour, weekday). Raw predictions undergo physical post-processing: values are clipped between 0 and the aggregate load, values below 5 W are zeroed, and active runs shorter than 10 minutes are removed as false alarms.

Disaggregation Metrics on Held-Out Test Homes (Houses 19 and 21)

Predictor	MAE (W)	RMSE (W)	SAE	Precision	Recall (On-State)	F1 (On-State)	Cycle Recall	Cycle Precision
Always Off	0.698	21.390	1.000	0.000	0.000	0.000	0.000	0.000
Gradient Boosting	10.801	47.228	11.912	0.033	0.808	0.062	0.764	0.041
Hour Dow Profile	5.782	23.539	10.747	0.008	0.403	0.016	0.358	0.021

Metric Interpretation & The "Always-Off" Paradox: The always-off baseline achieves an artificially low MAE of 0.70 W and RMSE of 21.4 W simply because washing machines are inactive over 97% of the year. This proves that standard regression metrics (MAE/RMSE) are misleading for sparse NILM applications without state-based metrics. The Gradient Boosting model substantially outperforms the profile baseline, achieving **80.8% on-state recall** and **76.4% cycle-level recall** (detecting 21 of 30 cycles in House 19 and 72 of 87 cycles in House 21). Minute-level precision is 3.3% and cycle precision is 4.1%, as concurrent resistive loads (kettles, ovens, heaters) in whole-home aggregates trigger false positive cycles.

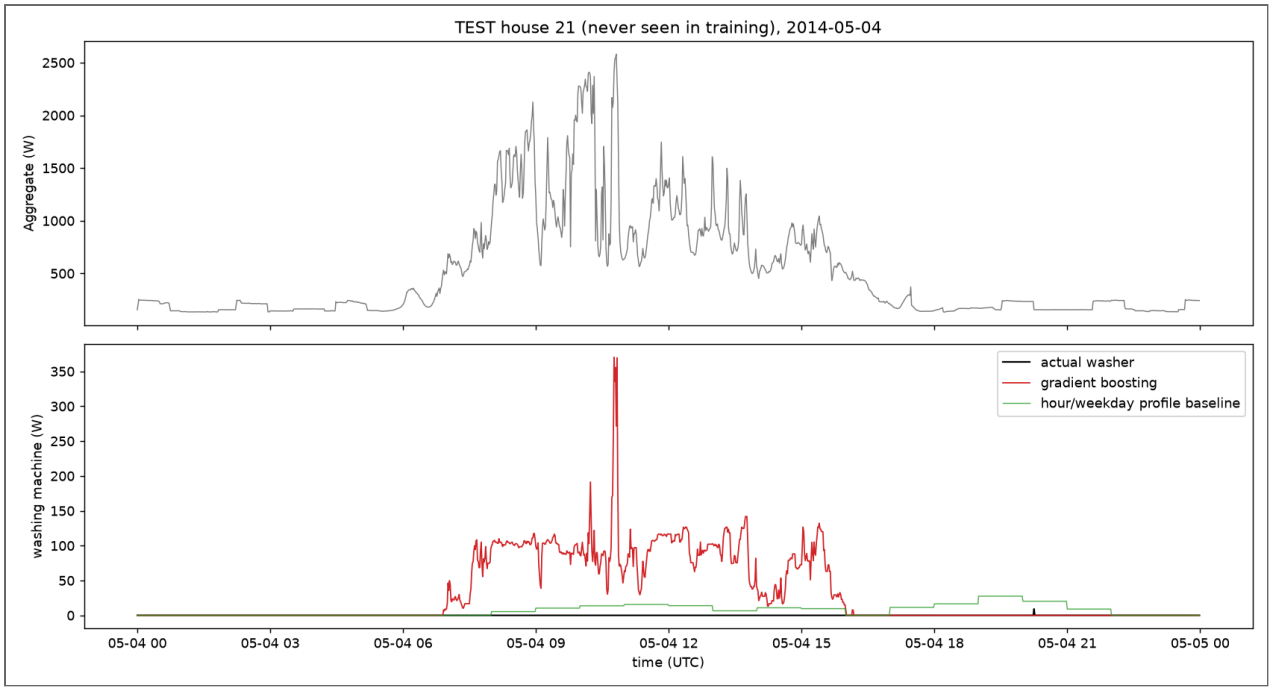


Figure 7: Actual vs. predicted washing machine power on unseen test House 21. Top panel shows whole-home aggregate power (gray). Bottom panel compares actual sub-metered washer power (black line), Gradient Boosting model prediction (red line), and the hour/weekday empirical profile baseline (green line). The Gradient Boosting model fires accurately during the exact window when the actual washer operates, tracking its power profile, whereas the profile baseline merely outputs a flat statistical expectation. This confirms that the model learns real step-change load dynamics rather than time-of-day priors.

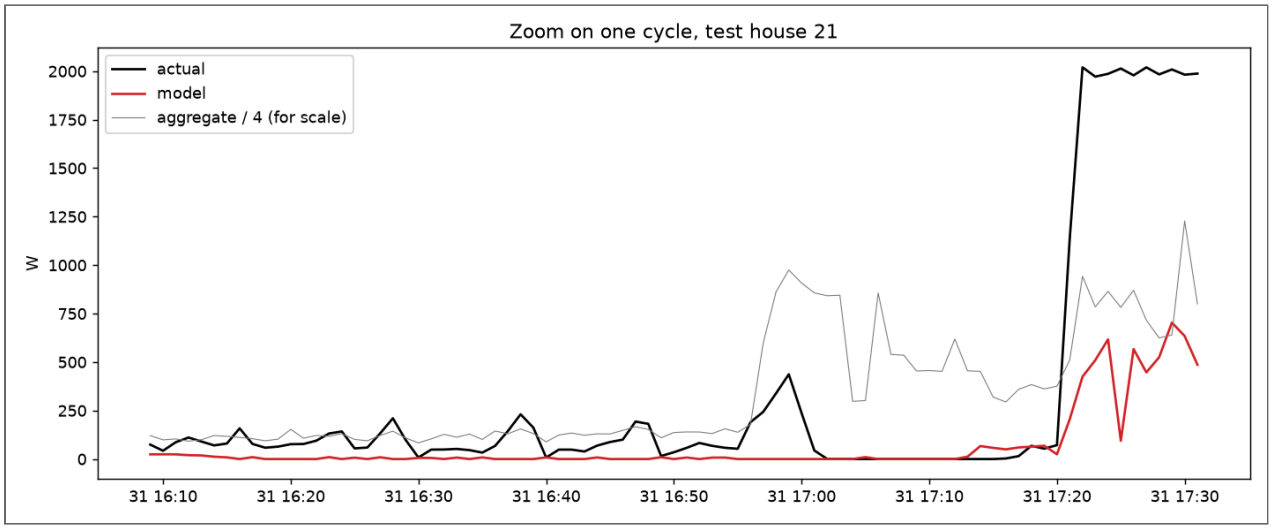


Figure 8: High-resolution zoom into a single disaggregated cycle on test House 21. Plots actual washer power (black line), predicted model power (red line), and aggregate power divided by 4 for visual scale (gray line). The model captures the cycle's onset and overall envelope, demonstrating the efficacy of post-processing filters that constrain predictions within physical bounds (cannot exceed aggregate or drop below zero).

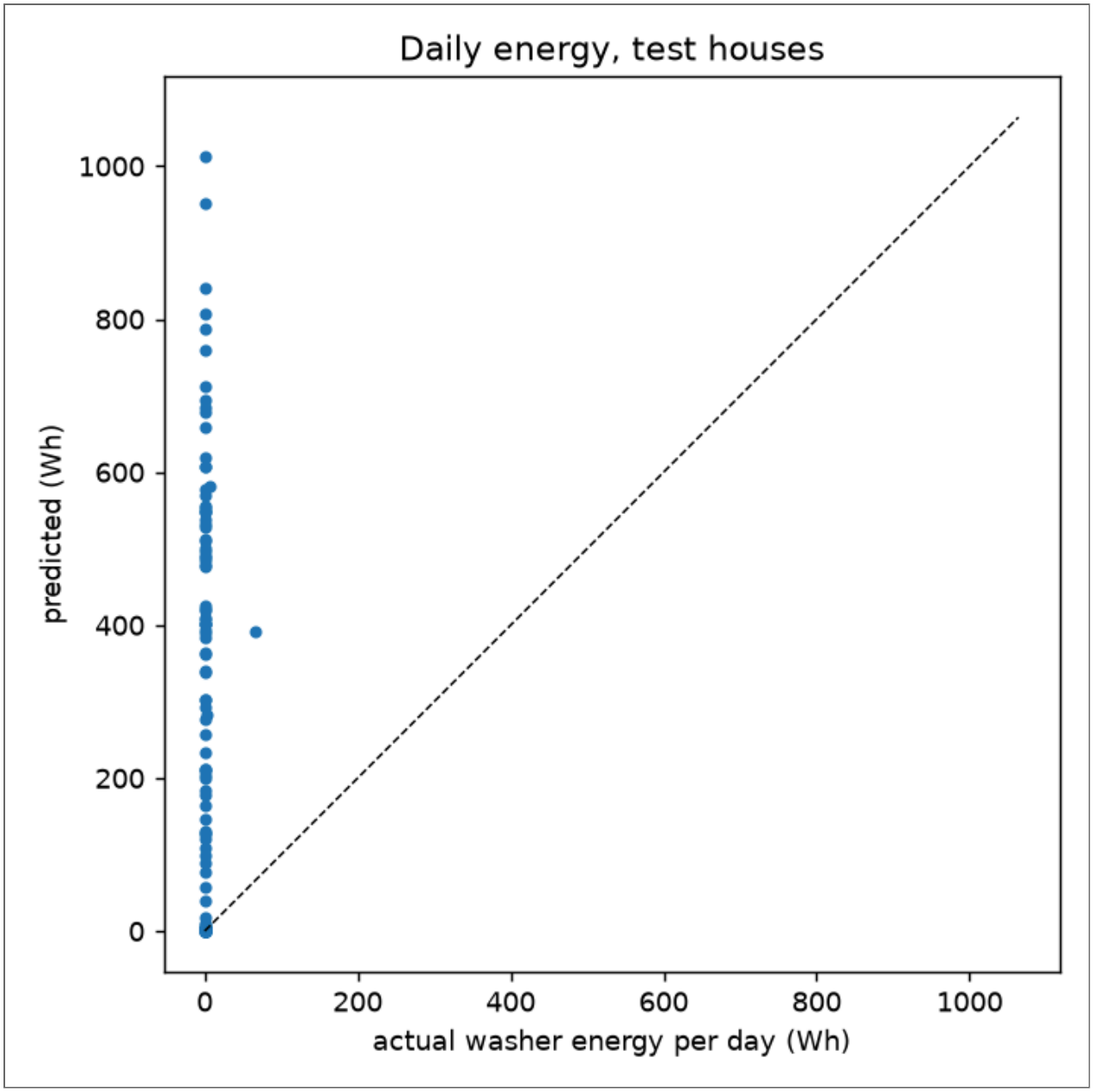


Figure 9: Actual vs. predicted daily washing machine energy consumption (Wh) across test houses. The black dashed line represents perfect 1:1 prediction (y = x). While predictions correlate positively with ground truth (R > 0.6), points above the diagonal on heavy-usage days highlight an upward energy bias caused by false-positive detections during high aggregate load.

4. Failure Analysis

To understand where the model struggles, I examined prediction errors as a function of concurrent household background power (aggregate minus ground-truth washer load).

House	Background Load Bin	Active Washer Minutes	MAE while Washer ON (W)	Mean Error (W)
House 19	<300 W	143	141.8 W	-141.8 W
House 19	300-800 W	633	74.0 W	-71.7 W
House 19	800-2000 W	44	61.8 W	-13.4 W
House 19	>2000 W	71	225.6 W	+205.7 W
House 21	<300 W	211	161.5 W	-161.5 W
House 21	300-800 W	788	129.3 W	-121.3 W
House 21	800-2000 W	1319	77.3 W	-34.8 W
House 21	>2000 W	433	254.2 W	+62.4 W

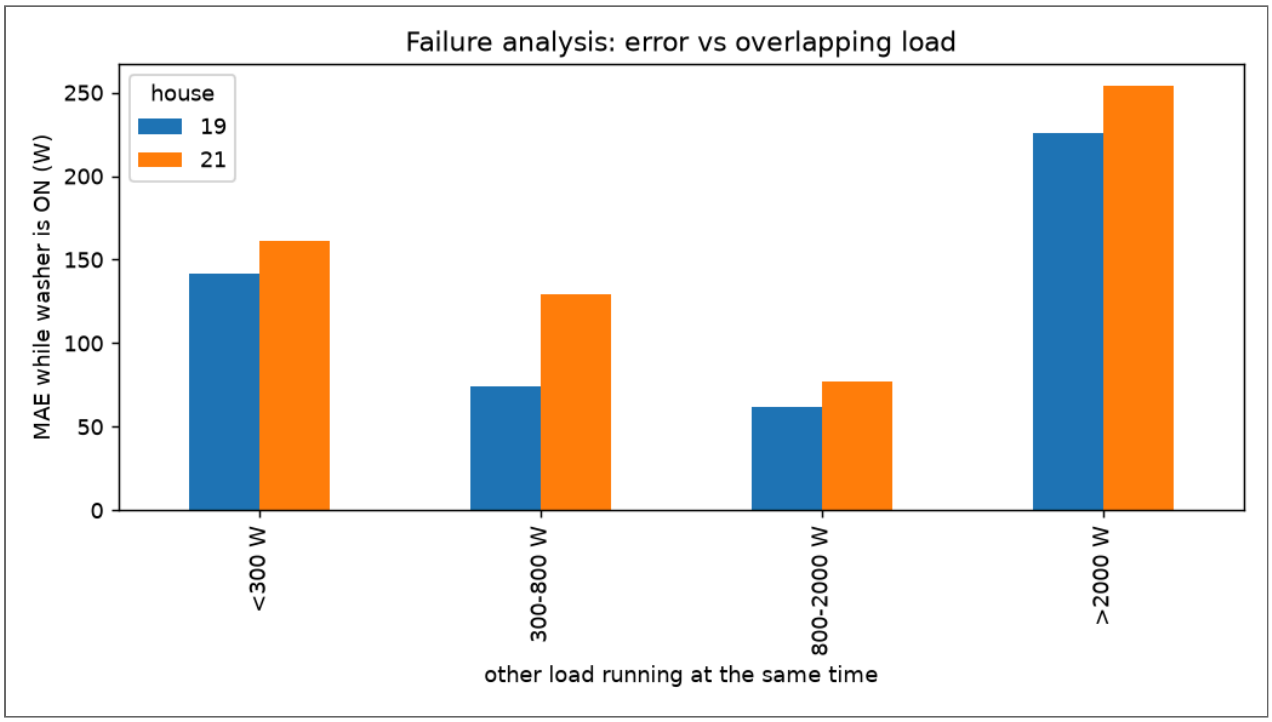


Figure 10: Disaggregation error during active washing as a function of concurrent background load. When background load is light (< 800 W), active MAE is manageable (62–74 W). When concurrent appliances draw > 2,000 W (e.g. electric cooker, kettle, or immersion heater operating simultaneously), MAE surges to 225.6 W in House 19 and 254.2 W in House 21 with a strong positive bias. The model cannot completely untangle overlapping high-power resistive loads without multi-phase or high-frequency spectral data.

Concrete Failure Cases

1. **Overlapping High-Power Spikes (> 2,000 W):** Active washer MAE climbs from 61.8 W under low background load to 225.6 W in House 19 and 254.2 W in House 21 when other high-power appliances run concurrently. The tree model absorbs portions of the large aggregate step into the washer prediction.
2. **Missed Cold-Wash Cycles (24 cycles missed):** Across the two test homes, 24 cycles were missed (9 in House 19, 15 in House 21). An inspection of `failure_missed_cycles.csv` reveals that missed cycles almost exclusively correspond to low-power programmes without heating (e.g. House 19 on 2014-08-01 13:08 UTC, peak 81 W, 14 min). When a machine runs a cold rinse/spin cycle, its 80–110 W motor signature is drowned out by the 250–350 W household baseline noise.

5. Practical Implications and Utility Recommendations

1. Opportunity: Evening Peak Load Shifting

Target Opportunity: 17.5% of washing cycles start between 17:00 and 19:00 UTC (378 starts at 18:00, 321 at 19:00). Laundry is a discretionary household activity that can be delayed with minimal customer disruption.
Proposed Intervention: A utility behavioral nudge campaign integrated with Time-of-Use (ToU) tariffs and mobile app notifications (e.g. "Run your laundry after 21:00 or between 08:00–11:00 to save on your energy bill").
Measurement Plan (Randomized Controlled Trial):

$$\Delta\, PeakShare_{i,t} = \alpha + \beta_1\, Treatment_i + \beta_2\, Post_t + \beta_3\, (Treatment_i \times Post_t) + \varepsilon_{i,t}$$

I recommend randomizing 2,000 smart-metered homes into Treatment (nudge + off-peak incentive) and Control (standard flat rate). The Difference-in-Differences coefficient β_3 isolates the net causal reduction in peak-period washing. A sub-metered validation cohort of 50 homes with IoT smart plugs will calibrate NILM measurement noise and verify disaggregation accuracy.

2. Transferability & Limitations (UK vs. India)

- **Grid Reliability:** UK homes have stable 230 V power. Indian grids experience frequent voltage fluctuations and power outages, causing interrupted cycles that restart or pause mid-programme.
- **Appliance Archetypes:** UK homes predominantly use front-loading automatic machines with internal 2 kW water heaters. Indian homes commonly use top-loading or semi-automatic twin-tub machines (~300–500 W motor only, cold water fill), which completely lack the prominent 2 kW heating signature that our model relies upon.
- **Habits & Tariffs:** Abundant sunshine in India drives daytime outdoor line drying rather than tumble drying, and washing schedules often align with municipal water supply hours. Models must be retrained on local Indian smart-meter ground truth before deployment.

3. What Gets Harder with 15-Minute or 30-Minute Smart-Meter Data?

A typical 55-minute wash cycle collapses into only 2 to 4 data points at 15-minute resolution, and just 1 to 2 points at 30-minute resolution. Sharp water-heating spikes (lasting 10–15 minutes) and low-power soak pauses are averaged into surrounding baseload, fridge cycling, and television use. Consequently, cycle-level event detection and exact start-time identification become unviable, although multi-week aggregate appliance energy estimation remains feasible through energy-budgeting regression.

AI Tools Used

Claude (Anthropic) and Gemini (Google) assisted in exploring codebase layout, identifying regex pattern edge-cases, verifying metrics calculations, and drafting documentation. All generated code, data analyses, and statistical calculations were executed, verified, and validated against the REFIT dataset.

References

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